



School of Computer Science
COMP 2067 Programming for Beginners
Summer 2025 – Lab 5
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Loops

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OBJECTIVES:

- i. To understand repetition structures.
- ii. To use while and/or for loops in a program.
- iii. To use break and/or continue instructions.

1. Launch **IDLE** in your computer, open a **new file** through IDLE, and name it ***lab5sol.py***.

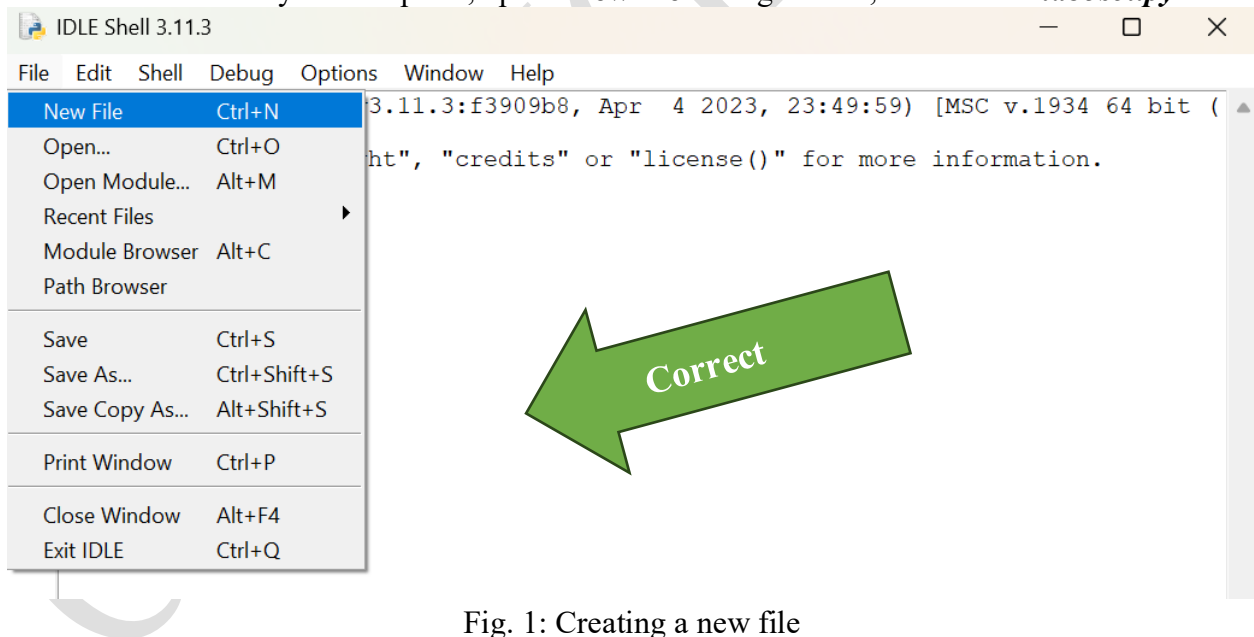


Fig. 1: Creating a new file

2. In *lab5sol.py*, write a Python program **with comments** to do the following:
 - a. Define a list of Canadian provinces and print it:
e.g. `provinces = ['Ontario', 'Quebec', 'Alberta', 'Manitoba', 'Saskatchewan', 'Nova Scotia', 'Newfoundland']`.
 - b. Create a variable ***sum_letters*** and initialize it to 0.
 - c. Create a for loop that iterates through each province in ***provinces***. For each province in the loop, do the following:

- i. Print the province name.
 - ii. If the first character of the province is 'A' (for Alberta), exit the loop with a suitable message¹. **HINT:** Use the break statement.
 - iii. Otherwise, add the number of characters in the province name to the current value of *sum_letters*.
 - d. After exiting the loop, print the total number of characters counted.
3. **Submit your Python code** (i.e., as lab5sol.py file) through **Brightspace**.

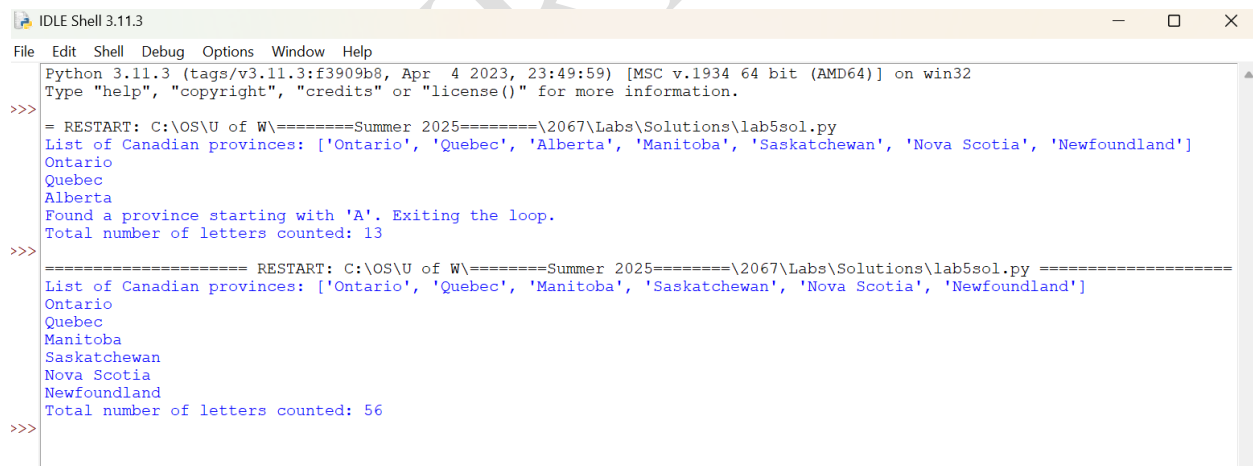
¹**NOTE:** In this course, the phrase 'suitable message' refers to some kind of description that describes what is being printed. For example, just printing the value **22** by itself does not tell the user anything useful; however, printing '**x = 22**' would add some context to what is being printed and so this would qualify as a 'suitable message' if the value of x is to be printed.

²**NOTE:** **step 2c** should be performed for all cases, i.e. it should be done whether the first character of the province is 'A' (for Alberta) or you don't have such a province.

³**NOTE:** **Do NOT submit a screenshot or the contents of the IDLE shell displaying your output. These will NOT be marked.** You must submit the *lab5sol.py* file.

Below is a **sample** screenshot showing example outputs for Lab 5.

Please note: do not submit a screenshot; only submit your lab5sol.py file. Your suitable message does not need to match the screenshot exactly.



```

IDLE Shell 3.11.3
File Edit Shell Debug Options Window Help
Python 3.11.3 (tags/v3.11.3:f3909b8, Apr 4 2023, 23:49:59) [MSC v.1934 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
= RESTART: C:\OS\U of W\=====Summer 2025=====2067\Labs\Solutions\lab5sol.py
List of Canadian provinces: ['Ontario', 'Quebec', 'Alberta', 'Manitoba', 'Saskatchewan', 'Nova Scotia', 'Newfoundland']
Ontario
Quebec
Alberta
Found a province starting with 'A'. Exiting the loop.
Total number of letters counted: 13
>>>
===== RESTART: C:\OS\U of W\=====Summer 2025=====2067\Labs\Solutions\lab5sol.py =====
List of Canadian provinces: ['Ontario', 'Quebec', 'Manitoba', 'Saskatchewan', 'Nova Scotia', 'Newfoundland']
Ontario
Quebec
Manitoba
Saskatchewan
Nova Scotia
Newfoundland
Total number of letters counted: 56
>>>

```

Fig. 2. A sample screenshot showing example outputs for Lab 5.

Submission:

- You will earn a maximum of 100 points (accounts for 2.5%) for submitting your **source code** (*lab5sol.py*) within the specified deadline.
- Do not upload your source code to Brightspace as a single zip/rar file.** Such submissions will not be accepted and a mark of **zero** will be assigned.
- Marks will be **deducted** if a *suitable message and/or comments* are missing.

4. Late submissions will incur a penalty of 10% per day for **up to two days**. Submissions made beyond two days will not be accepted, and a mark of **zero** will be assigned.
5. Unlimited resubmissions are allowed. But keep in mind that we will consider the last submission. That means that if you resubmit after the deadline, a penalty will be applied, even if you submitted an earlier version on time.

Academic Integrity.

Plagiarism is a serious offense and can result in severe consequences such as receiving a grade of **zero** on the assignment/lab or even being asked to leave the program.

Copying or using someone else's code is considered plagiarism. This includes using code from online sources, previous labs/assignments, or from other students. Even if you have modified the code, our antiplagiarism software can still show it as plagiarism.

To avoid plagiarism, make sure to always use your own words and ideas when writing source code. Additionally, always check with your instructor to make sure you understand what is allowed and what is not in terms of using outside resources.

Remember that academic integrity is essential for your own personal and professional growth, and it is your responsibility to uphold these principles. So please take it seriously and always produce your own original work.